The Binomial Expansion

Starters 2

1. Expand and simplify.

a $(2x-3y)^2$

a

b $(x-y)^3$

b

c $(2+x)^3$

c

2. Simplify.

a $(-2x)^3$

a

b $(3x)^{-4}$

b

c $(\frac{2}{5}x)^2$

c

d $(\frac{1}{3}x)^{-3}$

d

3. Simplify using root notation.

a $(25x)^{1/2}$

a

b $(64x)^{-2/3}$

b

c $(\frac{9}{100}x)^{-1/2}$

c

d $(\frac{8}{27}x)^{4/3}$

d

Chapter Overview 3

1. Pascal's triangle

Recognise the coefficients in binomial expansions.

2. Factorial and choose notation

Calculate coefficients directly.

3. Binomial expansion

Expand brackets and find a specified term.

4. Estimation

Use the first few terms to approximate powers.

Starter 4

0 Expand $(a+b)^0$

0

1 Expand $(a+b)^1$

1

2 Expand $(a+b)^2$

2

3 Expand $(a+b)^3$

3

4 Expand $(a+b)^4$

4

Coefficients The coefficients:

Coefficients

Powers The powers of a and b:

Powers

More on Pascal's Triangle 5

```
          1
        1   1
      1   2   1
    1   3   3   1
  1   4   6   4   1
1   5   10  10  5   1
```

Each interior entry is the sum of the two entries above it. Each row starts and ends with 1.

Number the top row 0. Row n supplies the coefficients of $(a+b)^n$. Learn rows 0 to 5.

Example Which row is used for $(2+x)^4$?

Example

Example 6

Find the expansion of $(2 + 3x)^4$.

Use descending powers of the first term and ascending powers of the second. The powers add to the power of the bracket. A term raised to power 0 is 1.

Coefficients First fill in the correct row of Pascal's triangle.

Coefficients

Term 1 | Term 2 | Term 3

Term 4 | Term 5

Result Simplify each term, raising every number in a bracket to the appropriate power.

Result

Initially write one line per term before simplifying. This gives each term enough space.

Another Example 7

Find the expansion of $(1-2x)^3$.

Use descending powers of the first term and ascending powers of the second. The powers add to the power of the bracket. A term raised to power 0 is 1.

Coefficients First fill in the correct row of Pascal's triangle.

Coefficients

Term 1 | Term 2 | Term 3

Term 4

Result Simplify each term, raising every number in a bracket to the appropriate power.

Result

$(1-2x) = (1+(-2x))$, so use $-2x$ as the second term. The signs alternate positive and negative; check this when simplifying.

Getting a single term in the expansion 8

The coefficient of x^2 in the expansion of $(2-cx)^5$ is 720. Find the possible values of the constant c.

The "5" row in Pascal's triangle is 1, 5, 10, 10, 5, 1. Counting the first entry as the 0th term, the coefficient we want is 10.

1 • The power of $(-cx)$ must be

1

2 • The power of 2 must be

2

3 Therefore the term is:

3

Test Your Understanding 9

a Find the first 3 terms, in ascending powers of x, of $(2 + kx)^7$, where k is a constant. Give each term in its simplest form.

a

b Given that the coefficient of x^2 is 6 times the coefficient of x, find k.

b

Exercise 8A – Y1 page 160-161 10

4 Fully expand the expression $(1 + 3x)(1 + 2x)^3$.

4

5 Expand $(2 + y)^3$. Hence or otherwise, write down the expansion of $(2 + x - x^2)^3$ in ascending powers of x.

5

6 The coefficient of x^2 in the expansion of $(2 + ax)^3$ is 54. Find the possible values of the constant a.

6

7 The coefficient of x^3 in the expansion of $(2 - x)(3 + bx)^3$ is 45. Find possible values of the constant b.

7

8 Work out the coefficient of x^2 in the expansion of $(p - 2x)^3$. Give your answer in terms of p.

8

9 After 5 years, the value of an investment of £500 at an interest rate of $X\%$ per annum is given by $500(1 + X/100)^5$

Find an approximation for this expression in the form $A + BX + CX^2$, where A, B and C are constants to be found. You can ignore higher powers of X.

9

Extension 13

MAT 2009 1J How many pairs of positive integers x,y satisfy

$x^3 + 6x^2y + 12xy^2 + 8y^3 = 2^{30}$

A: 0 B: 2^6 C: $2^9 - 1$ D: $2^{10} + 2$

MAT 2009 1J

Factorial and Choose Function 14

Factorial

$n! = n(n-1)\cdots 2 \cdot 1$

The number of arrangements of n distinct objects in a line. Also 0!=1.

For A,B,C, there are 3 choices, then 2, then 1:

$3! = 6$

Use your calculator's factorial function.

Choose

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

The number of selections of r objects from n distinct objects, when order does not matter.

Choose 4 players from 10:

$$\binom{10}{4} = \frac{10!}{4!6!} = 210$$

Use your calculator's nCr function.

Examples 15

Calculate the following. You may use the factorial button, but not the nCr button.

a $5!$

a

b $\binom{5}{3}$

b

c $0!$

c

d $\binom{20}{1}$

d

e $\binom{20}{0}$

e

f $\binom{20}{2}$

f

g $\binom{20}{18}$

g

Why do we care? 16

For large powers such as $(x+3)^{20}$, building Pascal's triangle becomes impractical. The choose function gives any entry directly.

n	row entries
0	1
1	1 1
2	1 2 1
3	1 3 3 1
4	1 4 6 4 1

$$\binom{4}{0} = 1, \quad \binom{4}{1} = 4, \quad \binom{4}{2} = 6$$
$$\binom{4}{3} = 4, \quad \binom{4}{4} = 1$$

Number the top row 0 and the first entry 0. Entry k in row n is $\binom{n}{k}$: the top number gives the row, the bottom number gives the entry.

The textbook starts counting at 1; here we count from 0 so the numbers match the binomial coefficient directly.

The triangle is symmetric:

$$\binom{4}{1} = \binom{4}{3}.$$

Choosing one object is equivalent to choosing the three objects to leave out.

Interesting facts (not required for A level) 17

Pascal's rule: each interior entry is the sum of the two above it.

$$\binom{3}{1} + \binom{3}{2} = 3 + 3 = 6 = \binom{4}{2}$$

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k}$$

To choose k objects from n, focus on one particular object. Either include it and choose k−1 from the other n−1, or exclude it and choose k from the other n−1. These cases are disjoint and cover every selection.

Exercise 8B – Y1 page 162 18

10 Given that $\binom{50}{13} = \frac{50!}{13!a!}$, write down the value of a.

10

11 Given that $\binom{35}{17} = \frac{35!}{p!18!}$, write down the value of p.

11

Exercise 8B – Y1 page 162 19

a Work out $\binom{10}{3}$ and $\binom{10}{7}$.

a

b Work out $\binom{14}{5}$ and $\binom{14}{9}$.

b

c What do you notice about your answers to parts a and b?

c

d Prove that $\binom{n}{r} = \binom{n}{n-r}$.

d

Using Binomial Coefficients to Expand 20

Why do binomial coefficients appear?

$(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b)$

Every product term chooses one item from each bracket.

Count How many selections produce a^3b^2?

Count

The whole coefficient row is:

$\binom{5}{0}, \binom{5}{1}, \binom{5}{2}, \binom{5}{3}, \binom{5}{4}, \binom{5}{5}$

$1, 5, 10, 10, 5, 1$

Using Binomial Coefficients to Expand 21

For a non-negative integer n:

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

The powers in each term sum to n. Negative and fractional powers require the Year 2 generalised expansion.

Example Find the first four terms of $(3x+1)^{10}$ in ascending powers of x.

Example

Test Your Understanding 22

Answer Find the first 3 terms in the expansion of $(2 - \frac{1}{3}x)^7$, in ascending powers of x.

Answer

Note: The "+ . . ." indicates that there would have been other terms in the expansion.

Exercise 8C – Y1 page 164

5 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(2-x)^6$ and simplify each term.

5

6 Find the first 3 terms, in ascending powers of x, of the binomial expansion of $(3-2x)^5$ giving each term in its simplest form.

6

7 Find the binomial expansion of $\left(x+\frac{1}{x}\right)^5$ giving each term in its simplest form.

7

Extension

1: AEA 2013 In $(1+\frac{12n}{5}x)^n$, the coefficients of x^2 and x^3 are equal and non-zero. Find n.

1: AEA 2013

2 [STEP I 2010 Q5a] By considering $(1+x)^n$, where n is a positive integer, or otherwise, show that:

$$\binom{n}{0}+\binom{n}{1}+\binom{n}{2}+\ldots+\binom{n}{n}=2^n.$$

2

Getting a single term in the expansion 25

For $(a+b)^n$ the term with b^r is

$$T_{r+1} = \binom{n}{r} a^{n-r} b^r$$

Write the bracket in an order that makes the required power easy to select.

1 Find the x^3 term in $(a + x)^{10}$.

1

2 Find the x^{50} term in $(2x - 1)^{75}$.

2

3 Find the x^7 term in $(3 - x)^{12}$.

3

4 Find the x^3 term in $(3x + 4)^{16}$.

4

Getting a single term in the expansion 26

The coefficient of x^4 in $(1 + qx)^{10}$ is 3360. Find the possible values of q.

1 Term is:

1

2 Therefore:

2

Test Your Understanding 27

Answer In $(1 + ax)^{10}$, where $a \neq 0$, the coefficient of x^3 is double the coefficient of x^2. Find a.

Answer

Exercise 8D – Y1 page 166-167 28

5. When $(1 - 2x)^p$ is expanded, the coefficient of x^2 is 40. Given that $p > 0$, use this informationto find:

a the value of the constant p

b the coefficient of x

c the coefficient of x^3

a Find p.

a

b Coefficient of x.

b

c Coefficient of x^3.

c

Question 6

a Find the first three terms, in ascending powers of x, of the binomial expansion of $(5 + px)^{30}$, where p is a non-zero constant.

a

b Given that in this expansion the coefficient of x^2 is 29 times the coefficient of x, work out the value of p.

b

Question 7

a Find the first four terms, in ascending powers of x, of the binomial expansion of $(1 + qx)^{10}$, where q is a non-zero constant.

a

b Given that in the expansion of $(1 + qx)^{10}$ the coefficient of x^3 is 108 times the coefficient of x, work out the value of q.

b

Exercise 8D – Y1 page 166-167 31

Question 8

a Find the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{11}$, where p is a constant.

a

b The first 3 terms in the same expansion are $1, 77x$ and qx^2, where q is a constant. Find the value of p and the value of q.

b

Exercise 8D – Y1 page 166-167 32

Question 9

a Write down the first three terms, in ascending powers of x, of the binomial expansion of $(1 + px)^{15}$, where p is a non-zero constant.

a

b Given that, in the expansion of $(1 + px)^{15}$, the coefficient of x is $-q$ and the coefficient of x^2 is $5q$, find the value of p and the value of q.

b

Exercise 8D – Y1 page 166-167 33

10 In the binomial expansion of $(1+x)^{30}$, the coefficients of x^9 and x^{10} are p and q respectively. Find the value of $\frac{q}{p}$.

10

Extension 34

1: MAT 2014 For positive integer n, find the coefficient of x^3y^5 in $(1+xy+y^2)^n$.

A: n B: 2^n C: $\binom{n}{3}\binom{n}{5}$

D: $4\binom{n}{4}$ E: $\binom{n}{8}$

1: MAT 2014

2: STEP I 2013 Compare coefficients of x^r in $(1+x)(1+x)^n$ to show

$$\binom{n}{r}+\binom{n}{r-1}=\binom{n+1}{r}$$

2: STEP I 2013

Estimating Powers 35

a Find the first 4 terms, in ascending powers of x, of $\left(1+\frac{x}{4}\right)^8$, giving each term in its simplest form.

a

b Use your expansion to estimate 1.025^8, giving your answer to 4 decimal places.

b

Why? Why should this be a reasonably good approximation despite the missing terms?

Why?

Tip: compare against the calculator value of 1.025^8.

Test Your Understanding 36

(a) Find the first 4 terms of the expansion of $\left(1+\frac{x}{2}\right)^{10}$ in ascending powers of x, giving each term in its simplest form. (4)

(b) Use your expansion to estimate the value of $(1.005)^{10}$, giving your answer to 5 decimal places. (3)

a

b

Exercise 8E – Y1 Page 168-169 37

7

a. Find the first 3 terms, in ascending powers of x, of $\left(3-\frac{x}{5}\right)^{10}$, giving each term in its simplest form.

b. Explain how you would use your expansion to estimate 2.98^{10}.

a

b

Exercise 8E – Y1 Page 168-169 38

8

a. Find the first 4 terms, in ascending powers of x, of $(1-3x)^5$.

b. Hence find a linear approximation to $(1+x)(1-3x)^5$ for small x.

a

b

9. A microchip company models the probability of having no faulty chips on a single production run as:

$P(\text{no fault}) = (1-p)^n, \quad p < 0.001,$

where p is the probability of a single chip being faulty, and n is the total number of chips produced.

a State why the model is restricted to small values of p.

a

b Given that $n = 200$, find an approximate expression for $P(\text{no fault})$ in the form $a + bp + cp^2$.

b

c The company wants a 92% likelihood of having no faulty chips on a production run of 200 chips. Use your answer to part b to suggest a maximum value of p for this to be the case.

c